

ADVANTAGE Section Discussion



Drilling Fluids

Operator	Peak Petroleum Indust. Limited	Field	Bilabri	Block/Co/Parish	OML - 122 / Nigeria	Facility	Bilabri DX 1
Well	Bilabri DX 1	Rig	BULFORD DOLPHIN	Job	Bilabri DX 1		

Section Details - 17 1/2" Section - #3

Mud Type	Water Base	Mud System	AQUA-DRILL	Top MD	459.00 m	Bot MD	1550.00 m
Start-End Date	07/Dec/2005-27/Dec/2005	Open Hole Diam	17 1/2 in	Total Days	20	Sect Length	1091.00 m
Section Comment	Drill 17 1/2" hole f/ 459 m to 1550 m. Run & cmt 13 13/8" csg						

Section Discussion

SECTION SUMMARY

Concentrated KCl brine was shipped from BHI base to be cut back to 12% by wt (average of 50:50 dilution with seawater) to be used for this interval. While RIH with 17-1/2" assy, 9.1 ppg Aquadrill mud was prepared.

Cmt was tagged at 444m & drilled out with seawater to 453m before displacing hole to AQUADRILL mud system. As mud arrived to surface, lined up shakers on mud & had initial shaker over flow with mud. Circulate to shear mud. At high flow rate under Brandt shaker box, observed over flow to over board ditch.

The mud was conditioned with polymers and seawater added for dilution while drilling. At 552m, pumped hi-vis sweep, circulate hole and POOH. While back on bottom commenced treating mud with KCl brine to increase potassium concentration in the active system. The formation was predominantly coarse sand and losses to the formation was observed. Hi vis pill with 15 - 30 ppb CaCO3 was pumped every stand drilled while treating the active system with CaCO3.

The sand traps were dumped at intervals. The Desander, Desilter & centrifuges were run while maintaining dilution with KCl premix into the active system to control solids & enhance inhibition. The hole was drilled to 1101m, swept with hi-vis pill & circulated clean before making a wiper trip to the 26" csg shoe. A maximum overpull of 40kips was observed on the trip & tight spots were reamed out. The hole was further reamed to btm without fill & drilling continued with pumping of sweeps at interval & dumping the sand traps at an average of every 3 stds drilled to control solids & create room for dilution. A maximum gas of 5% was observed at 13338m & a flow check confirmed hole to be static.

To further control solids build up & mud wt due to the predominantly very fine sand being drilled, the shaker screens were changed to 175s while putting all the solids control equipment - Desander/Desilter/Centrifuges on stream. The well was drilled to TD of 1550 where the formation was now observed to be shale. A wiper trip was freely done to the shoe & back to btm without drag or fill. The hole was circulated clean & 50 bbls hi-vis sweep spotted on btm before tripping for 13-3/8" csg run.

The 13-3/8" csg was run & got stuck differentially at 968m. The mud wt was reduced from 9.7 to 9.5 ppg by dilution with seawater. This did not help free the csg & a 250 bbls 9.5 ppg 8 lb/bbl Soltex pill was spotted around the stuck area. The csg was freed with a 260kips overpull. The csg was circulated & worked for a while with the wt stabilized at 9.5 ppg before continuing the csg run.

With the csg at 1355m, the csg hanger was made up but it was observed the hole was sticking on the string. Working the pipe with an overpull of 210k couldn't free the csg. The mud wt was subsequently reduced from 9.5 to 9.2 ppg by dilution with seawater. Removed hanger & pup joint & running tool. R/U ball on spider & csg circ tool on top drive. Pulled csg free with 720k over pull & work pipe. R/U ball on spiders elevator & engage circ tool on top 13-3/8" csg joint & circ/condition mud with 10 lb/bbl CaCO3. 150 bbls Soltex/CaCO3 pill was pumped & 120 bbls spotted in the annulus. A dummy connection was performed & no drag observed. The csg hanger was made up again & continued in hole with landing string.

The csg got stuck again at 1536m & had to be worked for several hours without success. Soltex pill was mixed & displaced into suspected stuck point. 300 bbls seawater was later pumped down hole and later another 450 bbls seawater all in the effort to reduce hydrostatic pressure & free the differentially stuck csg but not successful. The seawater was displaced & the mud wt was cut back on the fly from 9.2 to 9.1 ppg & then conditioned for enhanced lubricity.

75 bbls Parateq base oil was pumped & spotted across suspected stuck point & the csg worked while allowing to soak. This was later displaced out with 9.1 mud & seawater subsequently spotted around 968m.

As these attempts couldn't free the csg, the csg was cemented at that point, 1536m, about 6m from the hanging point and a csg patch operation was initiated.

The casing was cut & some joints with the hanger retrieved. A milling tool was run in hole to dress the csg stump & thereafter the hole swept clean with hi-vis pill before running in with the patch.

The first csg patch did not hold & the patch assy was pulled out to run a mill. The csg stump was redressed with the mill, the hole swept with hi-vis pill and the patch successfully performed. Pressure test of the patch & seal assy was found to be ok.

Preparations for the next 12-1/4" phase then went underway.